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Feasibility of Meta Quest 2 for tracking upper-limb movements under varying field of view conditions: A pilot study

Vanina Costa^{1,*}, Rodrigo Garcia-Carmona^{2,+}, Cristina Sánchez^{1,+}, Eloy Urendes¹, Ana Rojo¹, Gisela Aragón¹, Clara Beatriz Sanz-Morère³, and Rafael Raya¹

¹Department of Information Technologies, Institute of Technology, Universidad San Pablo-CEU, CEU Universities, Urbanización Montepríncipe, 28660 Boadilla del Monte, Spain

²Information Processing and Telecommunications Center (IPTC), Universidad Politécnica de Madrid, ETSI Telecomunicación, Avda. Complutense 30, Madrid, Spain

³BioRobotics Group, Center for Automation and Robotics, Spanish Research Council, Madrid, 28500, Spain

*vanina.costacortez@ceu.es

+these authors contributed equally to this work

ABSTRACT

Measuring upper-limb movement using accessible motion capture technologies is of growing interest in rehabilitation and human motion analysis. While virtual reality (VR) systems offer an attractive alternative to traditional motion capture (MoCap), their performance under realistic usage conditions remains insufficiently explored. In this pilot study, we provide a preliminary evaluation of the feasibility of the Meta Quest 2 as a motion tracking system by comparing wrist trajectories obtained from VR controllers with those measured using a Vicon optical motion capture system. Three healthy participants performed upper-limb movements under different head orientations, allowing the analysis of tracking performance when controllers were inside and outside the field of view (FOV) of the head-mounted display. Results show that trajectory similarity between systems is high when controllers remain within the FOV (correlation above 0.830 ± 0.138) but decreases when controllers fall outside the visible area (down to 0.548 ± 0.360). These findings highlight the strong dependence of inside-out VR tracking performance on FOV conditions. Overall, the Meta Quest 2 demonstrates feasible performance for capturing wrist trajectories during upper-limb movements, although tracking consistency is significantly affected by controller visibility. These preliminary results provide insight into the limitations of consumer VR systems for motion tracking in realistic scenarios and should be considered when applying such devices in applied contexts.

Introduction

In the context of human motion analysis, it is essential to analyze and characterize upper-limb movement patterns¹. A variety of methodologies have been developed to measure joint angles and range of motion (ROM) to properly assess movement performance. From the earliest systems, such as goniometers, to the most sophisticated motion capture (MoCap) systems, they enable the quantification and characterization of human movement. In this way, the movement analysis process can be guided or evaluated using input from the analysis of the measurements these instruments provide². The goniometer is widely used in clinical practice for measuring ROM^{3,4}. It is considered a straightforward, versatile, and, especially, user-friendly tool⁵. However, the goniometer is limited to measuring joint angles in single planes and static positions, making it unfeasible for use in dynamic movements. Furthermore, its reliability and accuracy show significant variations. For intra-rater reliability, intraclass correlation coefficients (ICCs) range from 0.76 to 0.94 for shoulder and elbow ROM^{4,5}. For inter-rater reliability of the same joints, ICCs range from 0.36 to 0.91^{3,5}.

MoCap is a technology that can capture human motion for applications such as kinematic analysis, postural control, and pose estimation^{6,7}. MoCap systems are currently gaining popularity in movement analysis and sports science because they provide accurate joint angle sequences and can visually display the subject's motion⁸. These systems are the most widely used instruments in research because of their high accuracy and versatility in both static and dynamic planar tasks. MoCap systems can be classified into inertial-based, optical motion capture (OMC), and fusion-based systems⁶.

Inertial-based systems are comprised of multi-axis accelerometers, magnetometers and gyroscope sensors that gather data used to determine the position and orientation of body segments^{8,9}. Medical applications are increasingly incorporating wearable inertial sensors since they are relatively inexpensive and can effectively monitor human motion in everyday settings^{2,5,9}. Due

to their ease of use, portability, and affordability, they are gaining popularity, even though their precision is surpassed by other MoCap technologies.

OMC MoCap systems use high-resolution camera arrays to reconstruct a 3D (three-dimensional) space. These cameras track markers (active or passive) attached to different anatomical parts of the human body. Vicon and OptiTrack are considered the gold standard for three-dimensional motion analysis^{2,10,11}. They typically serve as the benchmark for validating measurement tools due to their high precision. However, measuring the upper limb using these systems is still considered a challenge due to the complex human biomechanics of the shoulder, elbow, and wrist¹². OMC systems are generally stationary, expensive, and demand significant operational expertise. Therefore, they are rarely used in clinical practice and are mostly limited to research settings.

There are more affordable depth camera-based MoCap systems, such as Microsoft Kinect and Leap Motion¹³, that employ two cameras: one equipped with an infrared emitter and the other with a standard visible spectrum emitter. Both systems have been employed for estimating upper limb motion by tracking 3D motion without the need for markers^{11,14,15}. However, these devices can only observe the subject from one angle, a limitation that produces a significant loss of accuracy in body tracking when performing self-occluding postures, such as bending or bimanual operations.

Fusion-based systems arise in response to these limitations. These are low-cost MoCap systems that combine inertial sensors with cameras (with or without markers)⁶. Since fusion-based systems use two data sources, the limitations of both inertial-based and OMC systems can be addressed¹⁶.

Most recent virtual reality (VR) systems based on head-mounted displays (HMDs) use a type of fusion-based MoCap system. These VR systems are capable of positional and rotational tracking of the HMD as well as additional VR devices and/or trackers¹⁷. Commercially available HMDs typically use either a beacon-based or an inside-out tracking system. Beacon-based tracking relies on beacons (also called “base stations”) set up in the room, similar to the camera arrays in OMC systems. Inside-out tracking, in contrast, has the cameras mounted on the HMD itself, freeing the user from a complex setup at the cost of potentially lower tracking precision.

Several VR setups have been evaluated for rehabilitation purposes, comparing them to an OMC system^{17–25}. These devices have demonstrated remarkable accuracy and precision during various translations and rotations, as indicated by the results. This reveals a growing interest in using VR systems as measurement instruments²⁶. However, most studies focused on the evaluation of VR systems for tracking natural body motion have been limited to beacon-based tracking devices^{20,22–24,26}. Validation studies of inside-out tracking devices are usually focused exclusively on the HMD itself, excluding the controllers²⁵, or are carried out with the HMD positioned so the controllers are always inside the viewpoint of the HMD cameras^{17–19,21}. Furthermore, the studied devices are often mounted on articulated structures or robot arms rather than human subjects. The real-world usage conditions does not conform to these setups, since the subject is not always looking towards the controller nor performing such precise, robotic movements.

Measuring upper body motion in an objective manner is important for analyzing movement patterns^{2,27}. In this context, upper body movement is marked by significant variability and complexity, posing challenges due to the many degrees of freedom (DoFs) of the kinematic chain involved when performing functional tasks²⁸. This variability requires that, to properly assess the state of a subject, their motions must be captured as frequently and accurately as possible. To this end, a relatively inexpensive and easy-to-set-up device, such as an inside-out based VR system that can be used in home or uncontrolled environments, could be ideal. Moreover, a VR device could serve as a powerful remote movement monitoring or training platform, increasing user engagement through interactive and gamified exercises while providing objective information about movement performance over time²⁹.

In this [preliminary investigation](#), we evaluate the performance of the Meta Quest 2 inside-out VR system for tracking upper-limb movements, with a specific focus on how tracking [consistency](#) is affected by the relative position of the controllers with respect to the field of view (FOV) of the head-mounted display. Unlike previous validation studies conducted under controlled conditions, where controllers remain consistently visible, this work explicitly examines tracking performance under varying head orientations that may cause controllers to move in and out of the FOV, thus reflecting more realistic usage scenarios.

To this end, wrist trajectories obtained from the Meta Quest 2 controllers are compared against those measured using a Vicon optical motion capture system. By analyzing tracking [consistency](#) across different visibility conditions, this [pilot](#) study aims to characterize a key limitation of inside-out VR tracking systems that is often overlooked: their dependence on controller visibility during natural movements. This contribution provides practical insight into the [limitations](#) of consumer VR devices when used outside ideal laboratory setups.

Methods

The VR device precision was evaluated by observing the position of the wrist during two distinct upper limb actions performed by 3 healthy young subjects. Joint movement was analyzed simultaneously using the Vicon system and then compared. The

tracking performance of the VR device was tested by measuring the position of the wrist joint in two specific [fundamental single joint upper limb movements](#): shoulder flexo-extension (SFE) and elbow flexo-extension (EFE). The position of the subject's wrists joints was captured in real-time at the same time by the VR setup (using the Meta Quest 2 Touch controllers) and by the Vicon system. 3 different orientations of the HMD were tested so the differences in [tracking performance](#) when the controllers are inside and outside the FOV of the HMD can be compared.

Meta Quest 2 Hardware and Software

The Meta Quest 2 incorporates inside-out tracking technology which relies on integrated infrared cameras to accurately track the user's location and movements without requiring of external sensors. These built-in cameras, combined with the measurements captured by the inertial systems present in the HMD and both controllers make up the device's fusion-based MoCap system.

To develop a real-time body tracking system that can be used in an immersive VR experience, the tracking data provided by the Meta Quest 2 HMD and its controllers was used. The Meta Quest 2 Touch are considered the end-effectors of the upper limb kinematic chain, representing the wrist joint. This decision was made because end-effector movement is a widely used outcome measure in movement analysis and robotics^{30,31}. The popularization of the end-effector outcome stems from the premise that by knowing the position of the endpoint of an arbitrary 3-joint kinematic chain, the motion profiles of the rest of the joints can be adequately defined³². Therefore, those approaches that can adequately capture the motion of the end-effector could, in turn, accurately estimate the motion of the entire kinematic chain.

A custom VR application was developed in Unity 3D (v.2021.3.45f2), using the XR Interaction Toolkit (v.2.6.3). This application registers the HMD and controllers position and orientation at a sampling rate of 72 Hz. The full code of the VR application has been made available through an open-source license and can be freely downloaded from the GitHub code-sharing website.

Data Recording

This section provides a description of the setup of both the Meta Quest 2 and the Vicon system for the data recording sessions. The full data recorded is publicly available from the Zenodo website.

Vicon Setup

The OMC cameras of the Vicon system were calibrated at the beginning of each test. Particular attention was paid to the positioning of the Vicon markers. For body tracking during dynamic tasks, the markers for the upper body were placed as depicted in Figure 1, following the standard Plug-in-Gait Marker Placement³³. The Vicon system has a sampling rate of 100 Hz.

Meta Quest 2 Setup

A room-scale area setup was performed before all measuring sessions and kept unchanged for all subjects. The virtual floor was defined at the same height as the physical area floor. At the beginning of each subject's session, the VR application was calibrated according to their frontal positioning. Since both head markers (LBHD and RBHD, shown in Figure 1) are placed at the HMD, the Vicon's head midpoint forward vector is expected to correspond with the forward vector in the VR coordinate system (see Figure 2). To acquire the trajectories of the upper body, global positions data were stored for all axes of the wrists (the controllers).

Start of Data Recording

The data recording process in the Meta Quest 2 was started by the subject, by pressing a button on one of the controllers. For the Vicon system, the recording was started by a laboratory technician. The subject and the technician initiated their respective recordings following the same auditory cue.

Tracking Status

To determine whether controllers were inside or outside the HMD cameras' FOV, we evaluated the tracking-status information provided by the Meta Quest API. However, preliminary testing revealed that these internal flags were not sufficiently precise for the temporal segmentation required in this study, as the API maintained a "tracked" status even after controllers left the visible camera area. Consequently, *Inside FOV* and *Outside FOV* conditions were defined experimentally according to the participant's head orientation and the side of the body performing the movement.

Participants and Experimental Procedure

Three healthy young subjects (two men, one woman) aged between 20 to 40, with no cognitive or physical impairments, voluntarily participated in the pilot study. All methods were performed in accordance with the relevant guidelines and regulations. Ethical approval was obtained from the Research Ethics Committee of the San Pablo CEU University (561/21/53). Informed consent was obtained from all participants prior to participation.

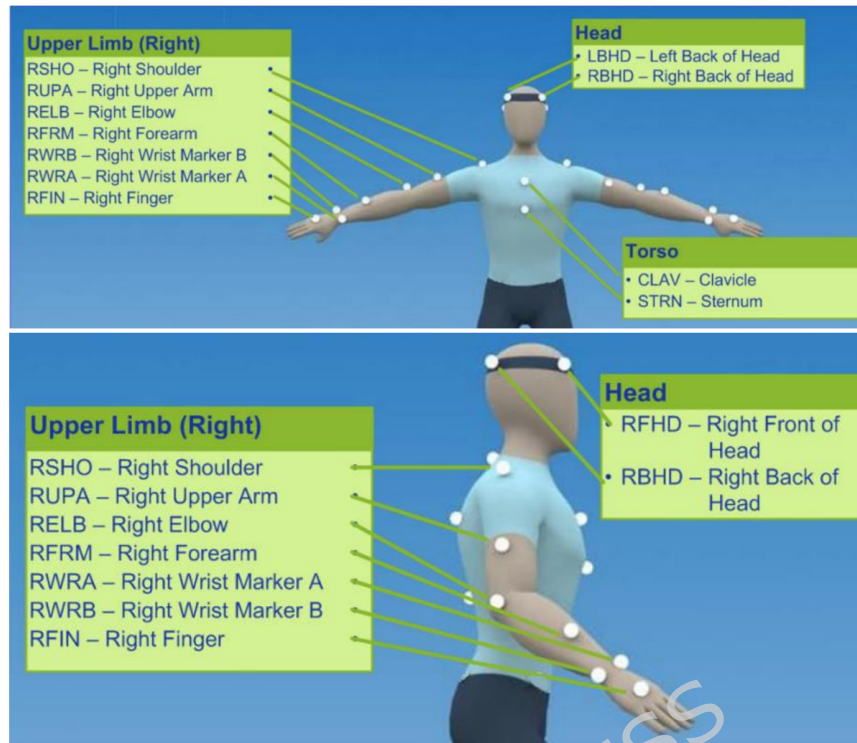


Figure 1. Front view of the attachment of Vicon markers to the upper body (top image); Right side view of the Vicon markers to the upper body (bottom image)³³.

During the execution of all motor tasks, the subject was seated on a chair without armrest, centrally positioned on a platform surrounded by the Vicon cameras. The subject holds one Meta Quest 2 Touch controller in each hand. The initial position for the motor tasks involves complete extension of the elbow, with the arm positioned close to the torso in a comfortable position and the wrist in a neutral position. The experiment comprises two movements in the sagittal plane, depicted in Figure 3:

1. *Elbow flexion-extension (EFE)*: The starting position corresponds to the arm being fully flexed in alignment with the torso (0°). Elbow flexion refers to the degree of convergence between the forearm and upper arm, reaching 135° . While elbow extension entails the opposite, with the forearm and upper arm moving away from each other. Normal active elbow flexo-extension ranges between 4.3° to 142.6° ³⁴.
2. *Shoulder flexion-extension (SFE)*: The starting position corresponds to the arm being fully flexed in alignment with the torso (0°). This movement involves raising the arm upward, keeping the elbow extension, in the sagittal plane from the initial zero position. Shoulder flexion corresponds to the degrees by which the arm moves in an upward direction, reaching 180° . While shoulder extension refers to the degrees in which the arm moves downward, returning closer to the zero position. Normal active shoulder flexo-extension ranges between -66.5° to 176.4° ³⁴.

The subject performed the designated movements with both the right and left arm at a self-selected speed three times. Additionally, the entire process is repeated three times, with each repetition involving a variation in the position of the subject's head. The HMD's orientation variations were done to assess whether the tracking performance of the system varies under different HMD facing positions. These variations include:

1. *Frontal orientation*: The subject's head is facing forward. In this position, the Meta Quest 2 Touch of both hands are within the field of view of the Meta Quest 2 HMD's cameras for all movements.
2. *Right orientation*: The subject's head is turned to the right. In this position, the right Meta Quest 2 Touch is within field of view of the Meta Quest 2 HMD's cameras for right SFE and EFE movements. However, it is not assured that the left Meta Quest 2 Touch is within the field of view of the Meta Quest 2 HMD's cameras for left SFE and EFE movements.
3. *Left orientation*: The subject's head is turned to the left. In this position, the left Meta Quest 2 Touch is within field of view of the Meta Quest 2 HMD's cameras for left SFE and EFE movements. However, it is not assured that the right Meta Quest 2 Touch is within the field of view of the Meta Quest 2 HMD's cameras for right SFE and EFE movements.

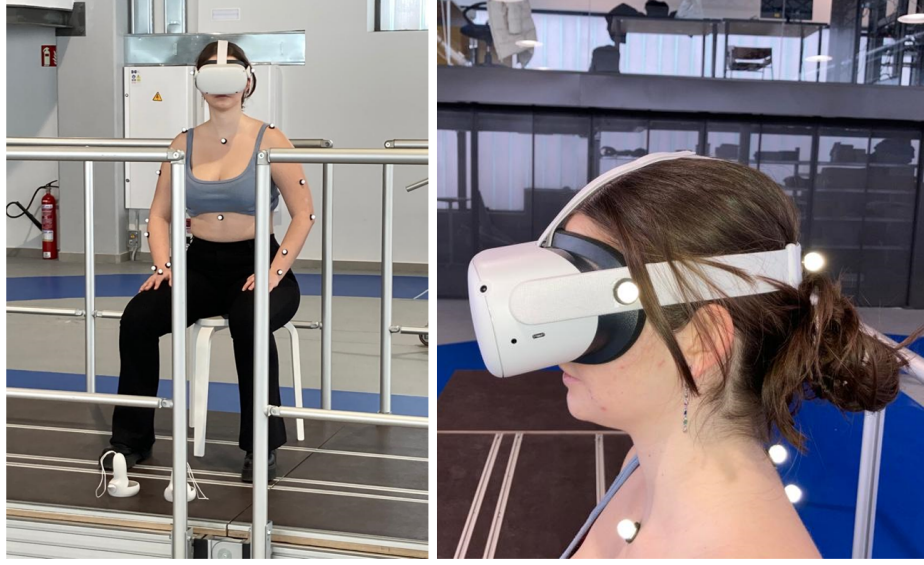


Figure 2. Preparation of the subject with the Vicon markers and the Meta Quest 2 HMD for the virtual environment calibration alignment of both coordinate systems.

Each subject performed 3 complete sets of the movements described for each arm and HMD positioning, with each set being the sequence of movement repetitions. Therefore, each subject performed 9 repetitions of the same movement per arm.

Data Processing

Wrist Data

The wrist joint centers were calculated from the Vicon raw positions by finding the midpoint of the wrist markers: LRA and LRB for the left wrist, and WRA and WRB for the right wrist (WRA and WRB shown in Figure 1). For the Meta Quest 2, the wrists joint centers are the Meta Quest 2 Touch positions.

Temporal Alignment of Signals

The signals were manually cut to ensure that the start of the Vicon recording began at time $t = 0$ and concluded at time $t = n$, corresponding to the exact start and end points of the Meta Quest 2 recording, respectively. This preprocessing procedure was applied to all movements for each participant. Because the Meta Quest 2 application recorded at 72 Hz while the Vicon system recorded at 100 Hz, trajectories from both systems were temporally aligned and resampled to obtain comparable time series for point-by-point analysis.

Although both systems recorded the same movements simultaneously, recordings were initiated independently following the same auditory cue and acquired at different sampling rates. Therefore, after manual trimming of the start and end points, temporal resampling was applied to obtain comparable time series for point-by-point analysis.

Alignment of Coordinate Systems

The raw data obtained from the experimental sessions required a preprocessing phase to address the misalignment of coordinate axes between the two systems. For this process MATLAB R2023a was used. The sagittal, transversal, and vertical axes correspond respectively to the global +X, +Y, +Z axes in Vicon and to the global +Z, -X, +Y axes in VR (see Figure 4). It is important to note that the transversal axis in the Vicon coordinate system is inverted compared to the transversal axis in the Meta Quest 2. Therefore, to align both systems, the transverse axis values from the Meta Quest 2 had to be inverted first.

To establish a common coordinate system, the first step was to select three points from both the Meta Quest 2 data (A) and the Vicon data (B). These three points were selected from the trajectory of the right wrist, and the homologous was done for the left wrist. These three position and rotation values for the Meta Quest 2 coordinate system were converted to the Vicon system. This transformation follows the criterion of minimizing least square errors. Then, to align both datasets, the Euclidean or rigid transform is applied following the equation: $R \times A + P = B$; where R is the rotation and P is the translation vector. If the data is noisy, it will minimize the least square error:

$$err = \sum_{i=1}^N ||RA^i + P - B^i||^2 \quad (1)$$

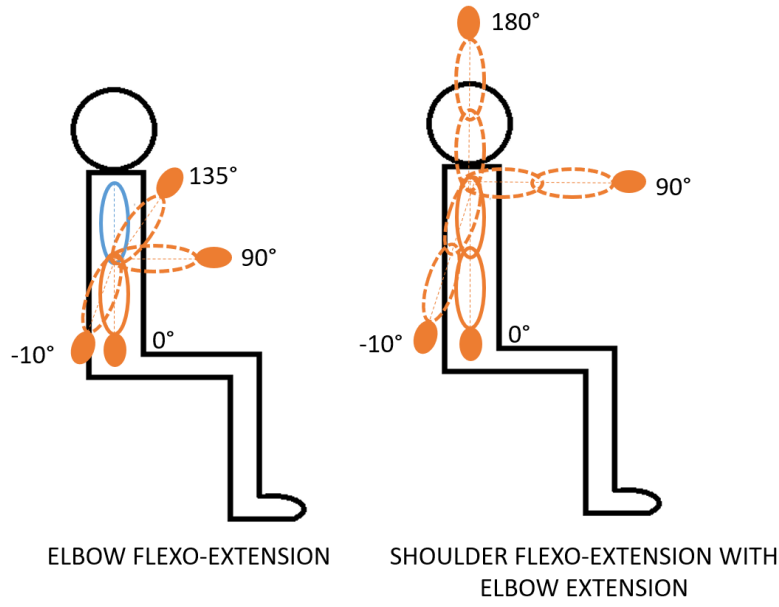


Figure 3. Diagram of elbow flexion-extension (on the left) and shoulder flexion-extension with extended elbow (on the right).

Then, to calculate the optimal rigid transformation matrix we determine the centroids of both data sets from the midpoints of the 3×1 vectors A and B , as expressed:

$$centroid_A = \frac{1/N^N}{\sum_{i=1}} A^i \quad (2)$$

$$centroid_B = \frac{1/N^N}{\sum_{i=1}} B^i \quad (3)$$

To determine the optimal rotation, Singular Value Decomposition (SVD) was employed so the three squared matrices could be obtained. Therefore, both datasets had to be re-centered to ensure that their respective centroids concurred with the origin (equations 4 and 5). Then, after, eliminating the translation component, the covariance matrix H is calculated (equation 6).

$$A_m = A - centroid_A \quad (4)$$

$$B_m = B - centroid_B \quad (5)$$

$$H = A_m \times B_m^T \quad (6)$$

Subsequently, both data sets are aligned to the origin. Finally, the optimal rotation matrix (R) and the translation vector (P) can be found applying SVD, as follows:

$$[U, S, V] = SVD(H) \quad (7)$$

$$R = V \times U^T \quad (8)$$

Where U is a $n \times k$ matrix with orthonormal columns, so that $U^T U = I_k$, where I_k is the $k \times k$ identity matrix and V is an orthonormal $k \times k$ matrix, so that $V^T = V^{-1}$. Lastly, to determine the translation vector P , the initial equation was referred to: $R \times A + P = B$, where P can be directly computed.

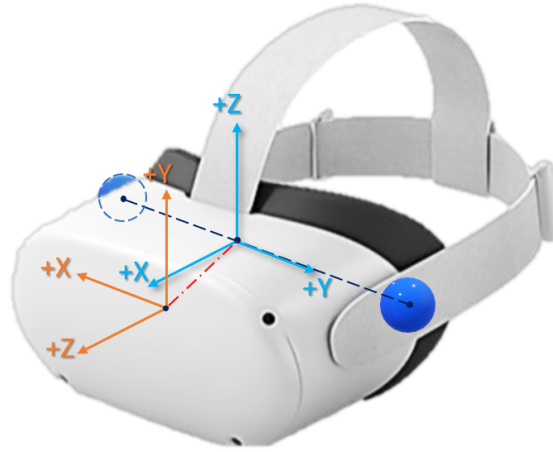


Figure 4. Misalignment of the Meta Quest 2 and Vicon coordinate systems: Blue axes represent the Vicon coordinates, orange axes represent Meta Quest 2 coordinates.

Data Analysis

A comparative analysis was conducted using Python 3.11 and the Statistical Parametric Mapping (SPM) library to obtain the mean and standard deviation over time for each participant and movement. Before analysis, the Meta Quest 2 and Vicon trajectories were expressed in a common coordinate system as described above. Since the two systems recorded at different sampling rates, the trajectories were temporally aligned and resampled to allow point-by-point comparison between systems and across repeated trials. For the descriptive trajectory plots, repeated trials were resampled to a common length before averaging. Mean trajectories and standard deviations were then computed over time for each participant, movement, arm, and HMD orientation. Trajectories were represented in metric units after coordinate-system alignment. To evaluate trajectory similarity between systems, Pearson correlation coefficients were calculated between the temporally aligned sagittal-plane wrist/controller trajectories obtained from the Meta Quest 2 and Vicon systems. The resulting coefficients were interpreted as measures of waveform similarity and tracking consistency over time, rather than as measures of absolute spatial agreement.

Results

This section begins with the distribution analysis results of the wrist motion during EFE and SFE movement in sagittal plane for the three orientations, including, as an example of visualization, the graphs for one of the three participants. Then, it shows the average results across subjects of the wrist movement curve of the EFE and SFE movements.

Overall Meta Quest 2 Performance: Descriptive Analysis

To analyze the data distribution, the average movement curve of a specific movement performed by the subjects using both systems was calculated. Graphs of the average movement curve and SD are shown for the elbow flexo-extension and the shoulder flexo-extension in sagittal plane. The data from the Meta Quest 2 are represented in blue, while the data from the Vicon system are represented in red.

The wrist translations for EFE and SFE movements of a specific subject are presented in Figure 5 and Figure 6, respectively.

Tracking consistency: Correlation Analysis

Pearson correlation coefficients between the trajectories recorded by the Meta Quest 2 and Vicon systems are reported for each movement condition.

Table 1 shows the overall correlation outcomes of all the subjects. These assessments aimed to determine the similarity between the wrist trajectories recorded by the Meta Quest 2 and those captured simultaneously by the Vicon system during SFE and EFE movements under different HMD orientations.

Table 2 shows the average correlations of the movements performed inside the field of view (FOV) of Meta Quest 2 cameras and outside the field of view of the Meta Quest 2 camera in orientations other than frontal. That is: Inside-FOV aggregates the correlation values of right orientation with right hand, and left orientation with left hand. While Outside-FOV aggregates the correlation values of right orientation with left hand, and left orientation with right hand. Correlation values are calculated following this grouping criterion to eliminate possible discrepancies observable in Table 1 between sides.

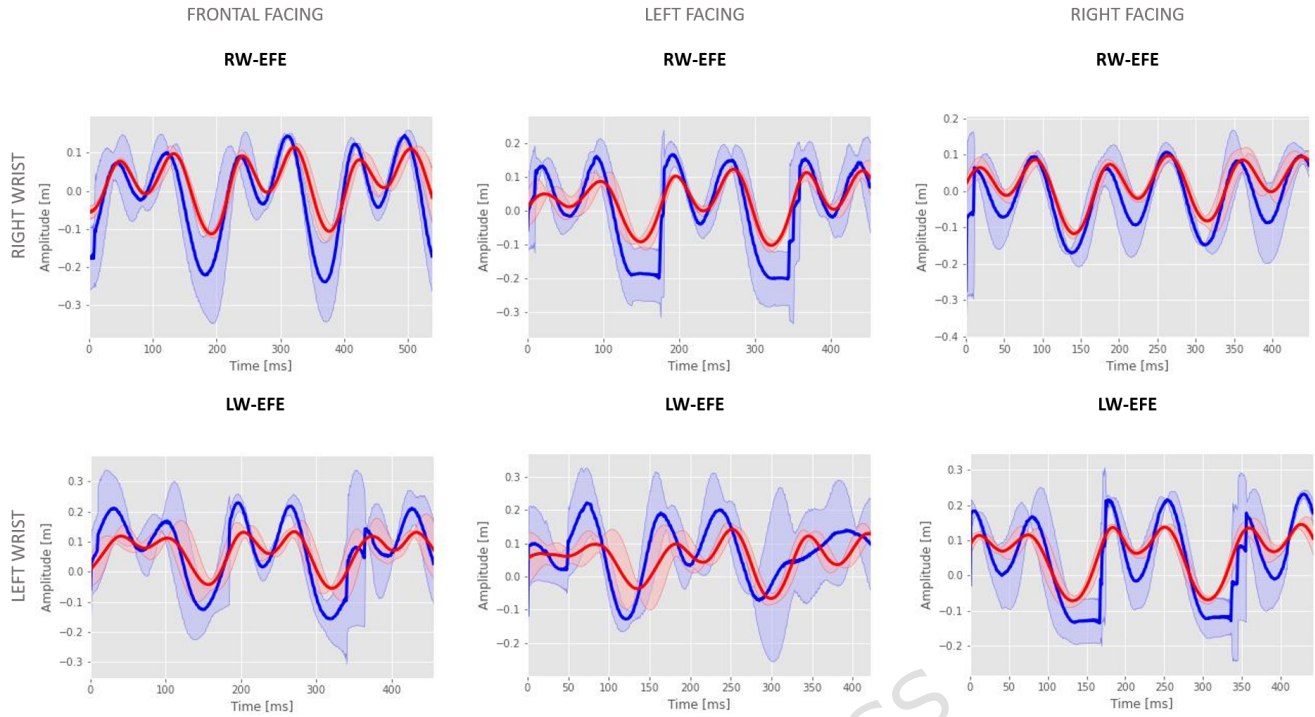


Figure 5. Each column (from left to right) shows the average **EFE** curve of the right wrist (RW) and left wrist (LW) in sagittal plane in frontal, left and right orientations. **Blue line represents the time-aligned average sagittal-plane wrist/controller trajectory captured by the Meta Quest 2; red line represents the corresponding trajectory captured by the Vicon system. Shaded areas represent variability across repeated trials. Amplitudes are shown in meters after coordinate-system alignment.**

Table 1. Correlation between the measurement of the Meta Quest 2 and the Vicon system. Wrist correlations (mean $\pm$ standard deviation) of overall movements (EFE and SFE) for the combinations of all HMD orientations (frontal, right, and left) and both arms (right and left) in sagittal plane of all participants.

HMD Orientation	Arm	Movement	Wrist Correl. (Mean $\pm$ SD)
Frontal	Right	EFE	0.944 ± 0.069
Frontal	Right	SFE	0.859 ± 0.114
Frontal	Left	EFE	0.834 ± 0.211
Frontal	Left	SFE	0.830 ± 0.138
Right	Right	EFE	0.891 ± 0.086
Right	Right	SFE	0.867 ± 0.091
Right	Left	EFE	0.700 ± 0.258
Right	Left	SFE	0.668 ± 0.291
Left	Right	EFE	0.621 ± 0.358
Left	Right	SFE	0.548 ± 0.360
Left	Left	EFE	0.709 ± 0.242
Left	Left	SFE	0.614 ± 0.266

Discussion

The aim of this work is to assess the feasibility of the Meta Quest 2 VR device as a MoCap instrument for dynamic upper-limb movements performed in a virtual environment. We wanted to find out if an inside-out VR device is a feasible tool for capturing wrist trajectories during upper-limb movements in conditions that approximate realistic usage scenarios. Our results suggest that the performance of inside-out VR tracking is not only device-dependent but strongly conditioned by controller visibility, a factor that becomes critical under realistic movement conditions and is often overlooked in previous validation studies.

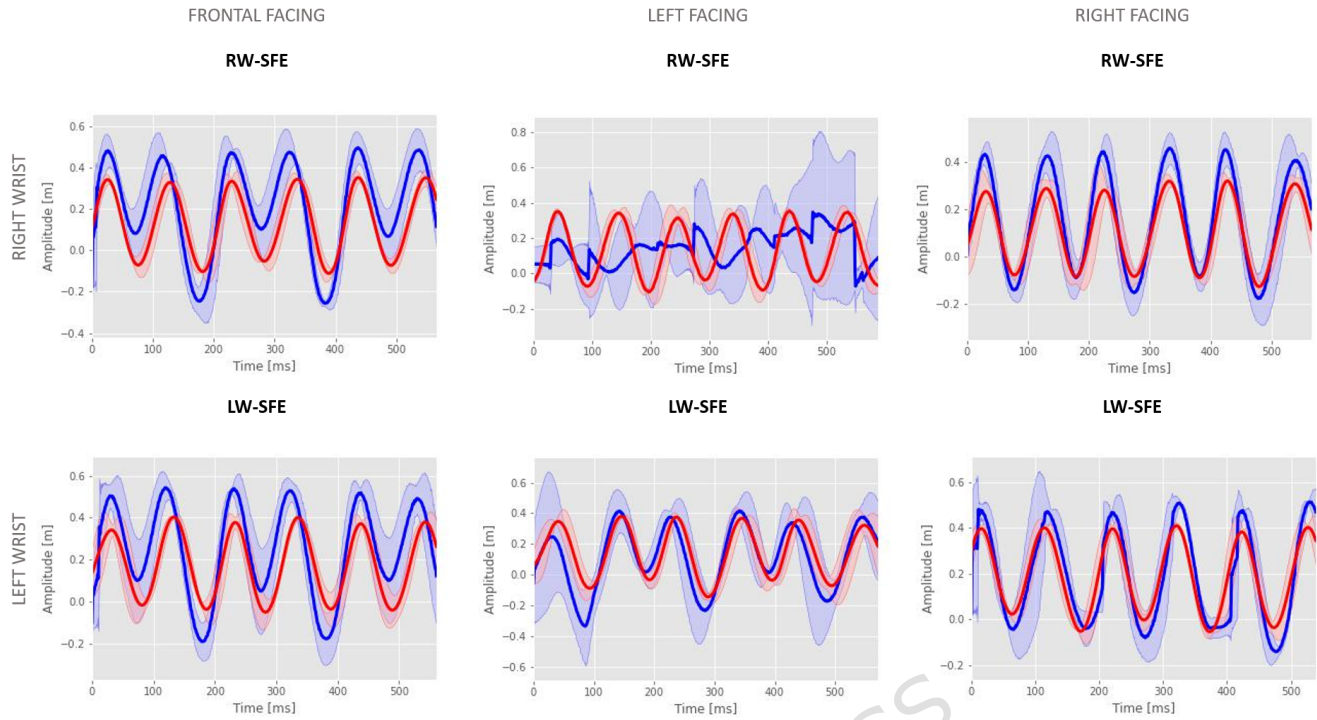


Figure 6. Each column (from left to right) shows the average SFE curve of the right wrist (RW) and left wrist (LW) in sagittal plane in frontal, left and right orientations. Blue line represents the time-aligned average sagittal-plane wrist/controller trajectory captured by the Meta Quest 2; red line represents the corresponding trajectory captured by the Vicon system. Shaded areas represent variability across repeated trials. Amplitudes are shown in meters after coordinate-system alignment.

Table 2. Correlation between the measurement of the Meta Quest 2 and the Vicon system. Wrist correlations (mean $\pm$ standard deviation) of overall movements (EFE and SFE) for “Inside-FOV” movements (right orientation with right hand, and left orientation with left hand), and “Outside-FOV” movements (right orientation with left hand, and left orientation with right hand) in sagittal plane of all participants.

Non-Frontal Inside/Outside FOV	Movement	Wrist Correl. (Mean $\pm$ SD)
Inside-FOV	EFE	0.800 ± 0.164
	SFE	0.741 ± 0.179
Outside-FOV	EFE	0.661 ± 0.308
	SFE	0.608 ± 0.326

Previous studies that have evaluated the accuracy of the Meta Quest 2 Touch controllers have been set up so the HMD always faced the controller directly. Carnevale, A. *et al.*¹⁷ studied the accuracy in terms of mean absolute error in translations and rotations of the controller versus the Qualisys system, obtaining a maximum mean absolute error of 13.52 ± 6.57 mm in translation and $1.11^\circ \pm 0.37^\circ$ in rotation. Similar results were obtained by Sum, L. C. *et al.*¹⁹, who reported a maximum mean absolute error of 3.5 ± 2.5 mm in translation. Pereira, D. *et al.*²¹ measured the accuracy for different brightness levels and in two different setups: one with the HMD placed on a horizontal surface and the other with a user wearing the HMD. They concluded that the mean squared error increased with rotational exercises, with a maximum mean error of 0.2° . This last study also reported that error values were more inconsistent in trials with subjects than when the HMD was fixed in place. The authors attributed those results to the participants moving their heads and making the controllers leave the field of view of the HMD. However, this specific situation is precisely what will happen in most real-world scenarios, since users immersed in a virtual environment will often find themselves in situations where the controllers move out of the HMD cameras’ FOV.

In this study, the comparison between systems focused on the similarity of the recorded wrist trajectories, using mean values to describe the overall movement patterns and standard deviations to characterize their variability. Correlation was used to assess similarity between the movement trajectories captured by both systems, reflecting how consistently the VR

device captures the overall patterns measured by the reference system. The results obtained in the descriptive analyses of the wrist show that the motions captured by the Meta Quest 2 and the Vicon systems show similar overall trajectory patterns. The morphology of the motion curves in Figures 5 and 6 shows that both curves describe a similar movement. However, it can also be seen that the Vicon system consistently captures a lower ROM. This systematic discrepancy could be due to the difference in sampling rates (72 Hz for the Meta Quest 2 and 100 Hz for the Vicon system) combined with the sensor fusion algorithm used by the Meta Quest 2. In fact, the infrared cameras of the Meta Quest 2 HMD physically sample at 50 Hz or 60 Hz, depending on the mains current of the country (50 Hz in our case). This observation could be especially important for works that measure the capabilities of VR devices to measure ROM³⁵.

The correlation analysis (Table 1) shows that the trajectory similarity between the Meta Quest 2 and Vicon systems remains relatively high in frontal orientation, generally above 0.830 ± 0.138 in frontal orientation. Tracking consistency in right and left orientations is significantly lower, ranging between 0.668 ± 0.291 and 0.891 ± 0.086 in right orientation, and between 0.548 ± 0.360 and 0.709 ± 0.242 in left orientation. This reduced tracking consistency in the non-frontal orientation movements contrasts with the absolute precision values reported in the literature, but coincides with the observation of Pereira, D. *et al.*²¹ that the values were more inconsistent when controllers moved out of the FOV of the HMD cameras. They also match the intuition that, when the controllers are out of the HMD inside-out cameras' FOV, the tracking performance decreases significantly.

In fact, our outcomes indicate that the consistency of the recorded trajectories is highly dependent on the orientation of the HMD. Table 2 shows the dependence on HMD orientation and controller position. For movements inside the FOV, the correlations are 0.741 ± 0.179 and 0.80 ± 0.164 for EFE and SFE, respectively; quite similar to the frontal orientation. However, for movements outside the FOV, the correlation drops to 0.661 ± 0.308 and 0.608 ± 0.326 for the EFE and SFE motions, respectively. These outcomes indicate that maintaining controller visibility within the HMD's FOV is important for consistent trajectory tracking. Therefore, the ability to maintain the controllers inside the FOV of the HMD has a big impact on the ability of an inside-out device to function as a motion capture system in applied movement analysis contexts. It should be noted that the reported correlations reflect trajectory similarity and tracking consistency between systems, rather than absolute positional agreement or interchangeability between tracking devices. The lower correlations observed outside the FOV should be interpreted as evidence of degraded tracking consistency rather than acceptable tracking performance. These values suggest that part of the overall movement pattern may still be preserved, but they may be insufficient for applications requiring precise quantitative motion analysis. In contrast, the higher correlations observed when the controllers remain within the HMD's FOV support the feasibility of the system for controlled or semi-controlled tasks in which controller visibility can be maintained.

It can also be seen that the correlation is lower in the SFE motion than in the EFE motion. We hypothesize that this is because both ends of the SFE motion probably fall outside the vertical FOV even in the frontal orientation (by virtue of being too high or too low), while in the EFE motion this only happens at one end (see Figure 3). This could also influence the different ROM values reported by the Meta Quest 2 compared with the Vicon system, since the ends of the motions determine the ROM. This reduction in tracking consistency could also be attributed to the wider ROM of the SFE motion.

It is important to note that, for both movements, the correlations (see Table 1) for the right motions performed while looking in the right direction (0.891 ± 0.086 and 0.867 ± 0.091) are significantly higher than those of the left motions performed while looking in the left direction (0.709 ± 0.242 and 0.614 ± 0.266). For the right orientation and right motion, the results are very similar to the frontal orientation and right motion (0.944 ± 0.069 and 0.859 ± 0.114). Meanwhile, for the left orientation and left motion, the results are very similar to the right orientation and left motion (0.700 ± 0.258 and 0.668 ± 0.291). This discrepancy is probably due to the differences in illumination on both sides of the room where the experiment was performed.

The Vicon system used was fixed in place in a room where outside light came through a set of windows, falling over the left side of the subject but not over their right side. Since both the Vicon system and the Meta Quest 2 cameras use infrared (IR) sensors, this type of daylight noise has a big impact on the quality of the results obtained by both tracking systems. In fact, OMC MoCap system vendors recommend that the room should be kept as dark as possible regarding IR light sources (like the sun) and reflective surfaces. Moreover, the technician in charge of the Vicon system noted that the data captured on the left side of the room are significantly noisier than the data obtained on the right side.

Furthermore, the potential influence of human anthropometry on tracking performance warrants consideration. Because inside-out tracking systems rely on cameras mounted on the HMD, individual anatomical differences may affect how consistently the controllers remain within the effective tracking volume during movement execution.

Also, Figures 5 and 6 show slight differences in ROM between the left and the right side. Side-to-side differences in range of motion have been previously reported in healthy populations and may be related to limb dominance and habitual movement patterns^{36,37}. These asymmetries could contribute to the differences observed between left and right facing orientations during the evaluated upper-limb movements. All participants were right-handed.

While the present study focused on upper-limb movements, the field-of-view limitations identified here may also have important implications for lower-limb tracking applications. During gait or seated lower-limb exercises, the lower extremities

may remain outside the HMD cameras' visible area for extended periods, potentially introducing additional tracking challenges.

These results highlight that the performance of inside-out VR tracking systems is strongly dependent on controller visibility, a factor that is often controlled or overlooked in previous validation studies.

Limitations

Several methodological limitations should be considered. First, the study included only three healthy participants and should therefore be interpreted as a preliminary pilot evaluation. In addition, wrist/controller position was used as a proxy for upper-limb movement, which does not capture joint-level kinematics. Future studies should include larger and more diverse samples and incorporate joint-specific outcome measures.

A further limitation is the use of relatively simple single-joint movements, which may not fully represent the complexity of Activities of Daily Living (ADL). More complex upper-limb tasks may introduce additional challenges for inside-out tracking systems, including self-occlusion and more variable movement trajectories. The simplified movements used in the present study were intentionally selected to isolate the influence of controller visibility and HMD FOV conditions on tracking performance.

Another limitation of the current study is the reliance on end-effector (controller) trajectories rather than joint-space kinematics. The Meta Quest 2 natively reports controller position and orientation data, whereas estimating joint angles would require inverse kinematics (IK) models that introduce additional modelling assumptions and potential sources of error. Therefore, the present study focused on wrist/controller trajectories to evaluate the native tracking output of the system under different field-of-view conditions. Additionally, absolute agreement metrics were not included, as positional differences between systems would also reflect the intrinsic offset between the handheld controller and the anatomical wrist reference estimated from Vicon markers.

When using the Meta Quest platform for motion capture, developers may choose between controller-based tracking and optical hand tracking. In the present study, controller-based tracking was selected because it represents a widely used interaction modality in VR rehabilitation applications and provides haptic feedback capabilities that may be beneficial in rehabilitation contexts. Optical hand tracking may present different sensitivities to FOV limitations and self-occlusion due to its vision-based tracking approach.

Additionally, environmental lighting conditions were not strictly controlled during the experimental trials and may have influenced the performance of the optical tracking system. However, the study was conducted under realistic indoor lighting conditions to better reflect potential home-based rehabilitation scenarios, where lighting environments are unlikely to be standardized.

Future Work

Future studies should evaluate the tracking performance of newer Meta Quest models, including the Meta Quest 3 and Meta Quest 3S. These devices provide a slightly larger combined camera field of view than the Meta Quest 2, although the camera placement differs between models. In the Meta Quest 3 and 3S, the lower camera placement may improve controller visibility near the waist or hips. In contrast, the upper corner camera arrangement of the Meta Quest 2 may provide better visibility during overhead movements, which are common in some rehabilitation exercises. This may make the Meta Quest 2 advantageous for specific movement scenarios despite being an older device. Additionally, the Meta Quest 3 incorporates dedicated IR illuminators, which may help reduce the influence of ambient infrared light under non-controlled lighting conditions.

Future work should also investigate the influence of anthropometric variability on controller visibility and tracking consistency across users with different body proportions. In addition, the feasibility of inside-out VR tracking systems for lower-limb movement analysis should be evaluated, particularly during gait and seated rehabilitation tasks in which the lower extremities may remain outside the HMD cameras' FOV for extended periods.

Conclusion

This study aimed to assess the feasibility of the Meta Quest 2 as a MoCap system for upper-limb motion tracking purposes by conducting a comparative analysis against the Vicon OMC MoCap system. The results of this research emphasize the crucial role that the orientation of the HMD plays in maintaining consistent controller tracking.

Nevertheless, the results suggest higher trajectory similarity when the controllers remain within the HMD's FOV, but substantially degraded tracking consistency when they move outside the FOV. Correlations as low as 0.548 ± 0.360 indicate important limitations of the system for applications requiring continuous and precise motion tracking. The highest correlation achieved (with the limbs performing the EFE motion inside the HMD's FOV) reached up to 0.944 ± 0.069 .

Future studies should evaluate the feasibility of capturing these movements in larger populations with upper-limb impairments and explore the applicability of these systems in relevant user populations.

Data availability

To support reproducibility and facilitate further research, we have made the following resources available:

- **VR application:** The VR application developed using Unity 2021.3.45f2 has its source code hosted on GitHub with an open source license, and can be downloaded freely: <https://github.com/rgarciacarmona/VR-Quest2-UpperLimb-MoCap>
- **Recorded data:** The data recorded following the procedures stated in the Methods section can be used freely: <https://doi.org/10.5281/zenodo.19234229>

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Author contributions statement

V.C., E.U. and R.R. designed the study; A.R. and R.G-C developed the software; G.A. and C.B.S-M. collected the data; C.S. and G.A. analyzed the data; C.S., R.G-C and A.R. interpreted the data; A.R., C.S. and R.G-C drafted the manuscript; V.C., E.U. and R.R. critically revised the manuscript. All authors reviewed and approved the final version of the manuscript.

Additional information

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